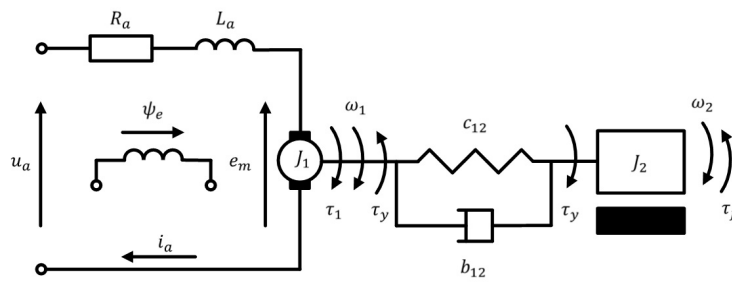


Computer Exercise: Disturbance Observer

Problem

Consider the drilling system that can be schematically represented as a two degrees-of-freedom (DoF) electromechanical system. Here, the first inertia J_1 is actuated by the torque τ_1 generated by the DC motor with separately excited field winding whereas a drilling (friction) torque τ_f is acting on the second mass.



The equations of motion are given for zero initial conditions of all system state variables in terms of Laplace variable s as:

$$\begin{aligned}\omega_1(s) &= \frac{1}{J_1 s} \left(\tau_1(s) - \tau_y(s) \right), \\ \tau_y(s) &= \frac{c_{12}}{s} \left(\omega_1(s) - \omega_2(s) \right) + b_{12} (\omega_1(s) - \omega_2(s)), \\ \omega_2(s) &= \frac{1}{J_2 s} \left(\tau_y(s) - \tau_f(s) \right), \\ \tau_1(s) &= \frac{\beta}{T_e s + 1} \left(\frac{1}{k_m \psi_e} u_a(s) - \omega_1(s) \right),\end{aligned}$$

where J_1 and J_2 are inertial masses of the drive system and the drill bit, $\omega_1(s)$ and $\omega_2(s)$ are angular velocities of two masses, c_{12} and b_{12} are torsional stiffness and damping of linear spring connection between masses, $\tau_1(s)$ is the drive torque, $\tau_y(s)$ is the elastic torque and $\tau_f(s)$ is the disturbance drilling torque, $u_a(s)$ is the armature voltage, $T_e = L_a/R_a$ is the electromagnetic time constant of the armature circuit, $\beta = (k_m \psi_e)^2 / R_a$ is the slope of the motor mechanical curve, k_m is DC motor ratio, ψ_e is constant magnetic field flow.

The control input to the system is the armature voltage, denoted as $u(s) = u_a(s)$. The disturbance affecting the system is the torque applied to the second mass, represented by $d(s) = \tau_f(s)$. The measured output is the angular velocity of the first mass, given by $\omega_1(s)$.

The parameters of the system are summarized in Table 1.

Parameter	Value	Description
R_a	0.3Ω	Armature resistance
L_a	10 mH	Armature inductance
T_e	L_a/R_a	Armature time constant
$k_m\psi_e$	0.085	Motor constant (for fixed magnetic flux)
β	$(k_m\psi_e)^2/R_a$	Slope of motor mechanical curve
J_1	$0.01 \text{ kg} \cdot \text{m}^2$	Inertia of first mass
J_2	$0.02 \text{ kg} \cdot \text{m}^2$	Inertia of second mass
c_{12}	0.1 Nm/rad	Spring coefficient between masses
b_{12}	$2 \cdot 10^{-4} \text{ Nms/rad}$	Damping coefficient between masses

Table 1: Parameters of the 2DOF Electromechanical System

Tasks

- Implement the plant in Simulink using the standard blocks. Neglect the disturbance input $d(t)$, simulate the step response of the system for the voltage amplitude 24 V .
- Derive a Luenberger observer (with the desired eigenvalues at locations -50 , -2 , -3 , -4) based on the state-space model of the system with the state vector

$$x(t) = \begin{bmatrix} \Delta\phi(t) \\ \omega_1(t) \\ \omega_2(t) \\ \tau_1(t) \end{bmatrix},$$

where the angle difference is defined as $\Delta\phi(t) = \phi_1(t) - \phi_2(t)$.

Implement the observer in Simulink and test it using the observer initial condition

$$\hat{x}_0 = \begin{bmatrix} 0 \\ 100 \\ 100 \\ 0 \end{bmatrix}.$$

- Apply disturbance to the plant. Analyze how different disturbances affect the estimated states.
hint:
 - to model the disturbance signals use the following blocks: Pulse generator (with Pulse width 1% of the period), Step, Sine wave.
 - take the amplitude of 1 Nm .
- Assuming the harmonic disturbance with frequency of oscillations 0.5 rad/s , design a disturbance observer to reconstruct the system state and disturbance.
- Analyze how measurement noise affects the observed states.